

COMBINATIONAL LOGIC DESIGN

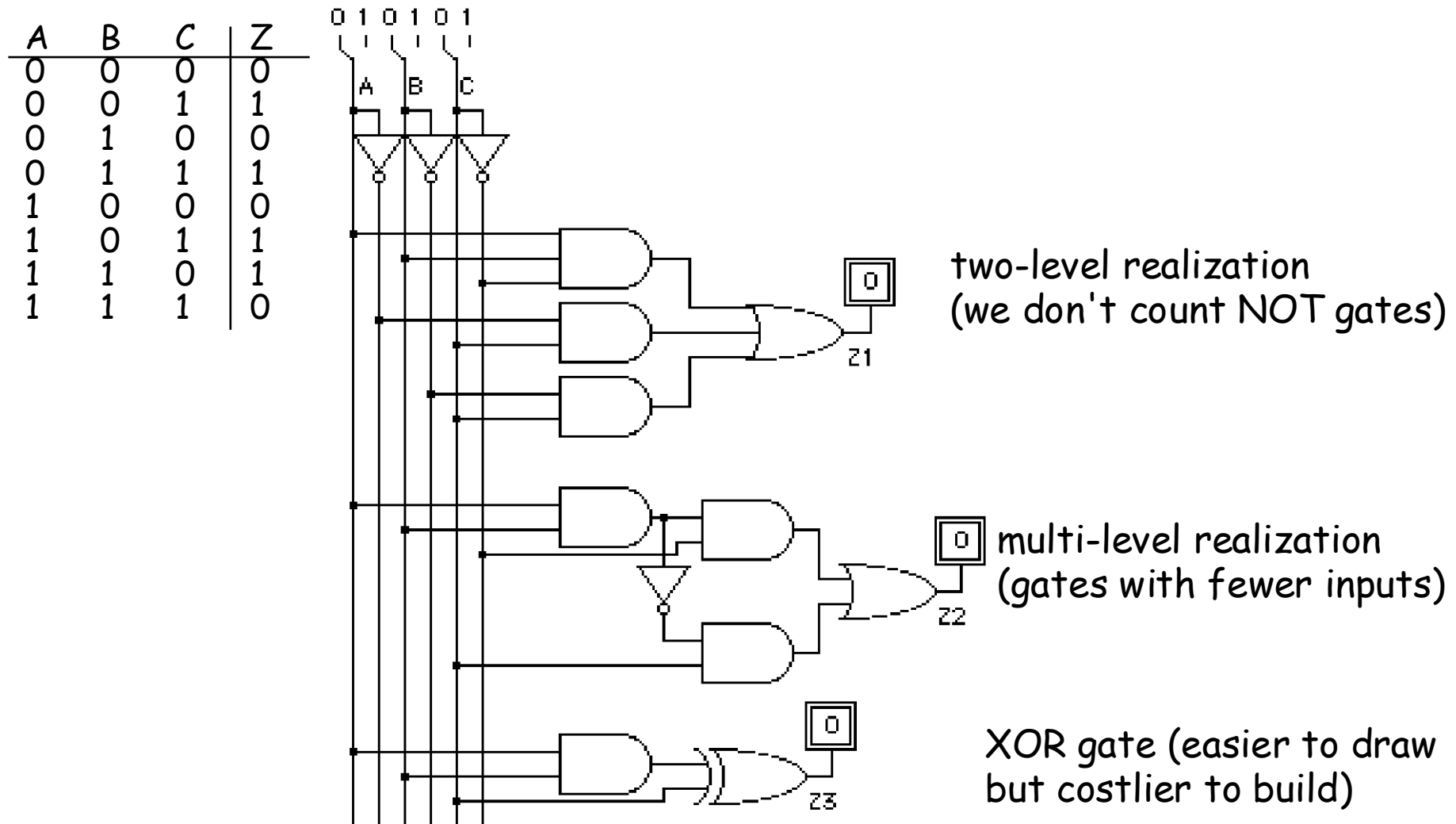
Module.2

CDA 3103C

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Different realizations of a function

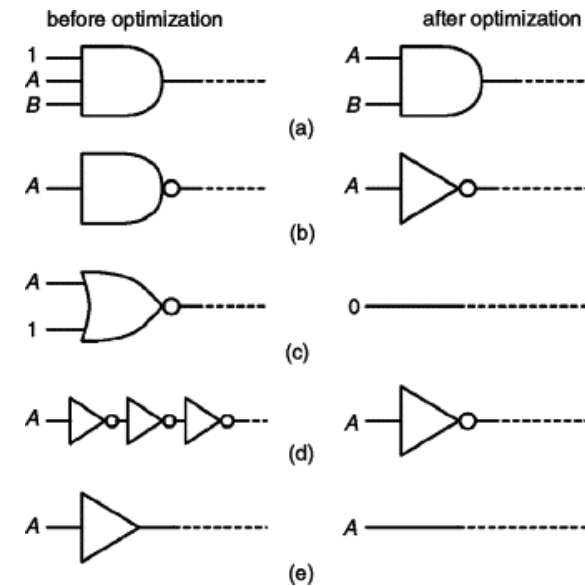


Which realization is best?

- Reduce number of inputs
 - literal: input variable (complemented or not)
 - can approximate cost of logic gate as 2 transistors per literal
 - why not count inverters?
 - Fewer literals means less transistors
 - smaller circuits and reduced electric connections
 - Fewer inputs implies faster gates
 - gates are smaller and thus also faster
 - Fan-ins (# of gate inputs) are limited in some technologies
- Reduce number of gates
 - Fewer gates (and the packages they come in) means smaller circuits
 - directly influences manufacturing costs

Which realization is best?

- Reduce number of levels of gates
 - Fewer level of gates implies reduced signal propagation delays.
 - Minimum delay configuration typically requires more gates.
 - wider, less deep circuits.
- How do we explore tradeoffs between increased circuit delay and size?
 - Automated tools to generate different solutions.
 - Logic minimization: reduce number of gates and complexity.
 - $PA + PA' = P(A + A') = P * 1 = P$
 - Logic optimization: reduction while trading off against delay.



Canonical forms

- Truth table is the unique signature of a Boolean function
- Many alternative gate realizations may have the same truth table
- Canonical forms
 - Standard forms for a Boolean expression
 - Provides a unique algebraic signature

Canonical forms

- Any logic function can be implemented with only AND, OR, and NOT functions.
- Any logic function can be written in canonical form, where every input is either a true or complemented variable and there are only two levels of gates
 - AND and OR

Canonical forms

- These are called two-level representations
 - sum of products (SOP)
 - A logical sum (OR) of products (terms using the AND operator)
 - product of sums (POS)
 - A logical product (AND) of sums (terms using the OR operator)

Note:

- ABC' product term
- $A+B+C'$ sum term
- $A'B + BC + AC'$ sum of products
- $(A + B + C)(A' + B' + C')$ product of sums

Logic functions in Canonical Form

- D is true if at least one input is true
- E is true if exactly two inputs are true
- F is true only if all three inputs are true.

Inputs			Outputs		
A	B	C	D	E	F
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	0	1

- $D = A+B+C$ Sum Term
- $F = A*B*C$ Product Term
- $E = ((A * B) + (A * C) + (B * C)) * (A * B * C)'$ Non-canonical
- $E = (A * B * C') + (A * C * B') + (B * C * A')$ Sum of Products

Sum-of-products canonical form

- Also known as disjunctive normal form
- Also known as **minterm** expansion

$F = 001 \quad 011 \quad 101 \quad 110 \quad 111$
 $F = A'B'C + A'BC + AB'C + ABC' + ABC$

A	B	C	F	F'
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

$F' = A'B'C' + A'BC' + AB'C'$

e.g., $F(A,B,C) = \sum m(1,3,5,6,7)$

$$F'(A,B,C) = \Sigma m(0,2,4)$$

Sum-of-products canonical form

- Product term (or minterm)
 - ANDed product of literals – input combination for which output is true
 - Each variable appears exactly once, in true or inverted form (but not both)

A	B	C	minterms
0	0	0	$A'B'C'$ m0
0	0	1	$A'B'C$ m1
0	1	0	$A'BC'$ m2
0	1	1	$A'BC$ m3
1	0	0	$AB'C'$ m4
1	0	1	$AB'C$ m5
1	1	0	ABC' m6
1	1	1	ABC m7

short-hand notation for
minterms of 3 variables



F in canonical form:

$$\begin{aligned}F(A, B, C) &= \Sigma m(1,3,5,6,7) \\&= m1 + m3 + m5 + m6 + m7 \\&= A'B'C + A'BC + AB'C + ABC' + ABC\end{aligned}$$

canonical form $\neq$ minimal form

$$\begin{aligned}F(A, B, C) &= A'B'C + A'BC + AB'C + ABC + ABC' \\ \text{Distributive} &= (A'B' + A'B + AB' + AB)C + ABC' \\ \text{Distributive} &= ((A' + A)(B' + B))C + ABC' \\ \text{Inverse, Identity} &= C + ABC' \\ \text{Commutative} &= ABC' + C \\ \text{Absorption} &= AB + C\end{aligned}$$

Sum-of-products canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Sum-of-products canonical form

Inputs			Output
A	B	C	D
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- $A' * B' * C$

Sum-of-products canonical form

Inputs			Output
A	B	C	D
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- $A' * B' * C$
- $A' * B * C'$

Sum-of-products canonical form

Inputs			Output
A	B	C	D
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- $A' * B' * C$
- $A' * B * C'$
- $A * B' * C'$

Sum-of-products canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

- $A' * B' * C$
- $A' * B * C'$
- $A * B' * C'$
- $A * B * C$

Sum-of-products canonical form

Inputs			Output
A	B	C	D
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- $$D = (A' * B' * C) + (A' * B * C') + (A * B' * C') + (A * B * C)$$

Product-of-sums canonical form

- Also known as conjunctive normal form
- Also known as **maxterm** expansion

			F =		F' =	
A	B	C	F	F'		
0	0	0	0	1	000	(A + B + C)
0	0	1	1	0	010	(A + B' + C)
0	1	0	0	1	100	(A' + B + C)
0	1	1	1	0		
1	0	0	0	1		
1	0	1	1	0		
1	1	0	1	0		
1	1	1	1	0		

$F' = (A + B + C') (A + B' + C') (A' + B + C') (A' + B' + C) (A' + B' + C')$

e.g., $F(A,B,C) = \Pi M(0,2,4)$

$F'(A,B,C) = \Pi M(1,3,5,6,7)$

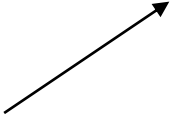
Product-of-sums canonical form

- Sum term (or maxterm)

- ORed sum of literals – input combination for which output is false
- each variable appears exactly once, in true or inverted form (but not both)

A	B	C	maxterms	
0	0	0	$A+B+C$	M0
0	0	1	$A+B+C'$	M1
0	1	0	$A+B'+C$	M2
0	1	1	$A+B'+C'$	M3
1	0	0	$A'+B+C$	M4
1	0	1	$A'+B+C'$	M5
1	1	0	$A'+B'+C$	M6
1	1	1	$A'+B'+C'$	M7

short-hand notation for
maxterms of 3 variables



F in canonical form:

$$\begin{aligned}F(A, B, C) &= \Pi M(0, 2, 4) \\&= M0 \cdot M2 \cdot M4 \\&= (A + B + C)(A + B' + C)(A' + B + C)\end{aligned}$$

canonical form $\neq$ minimal form

$$\begin{aligned}F(A, B, C) &= (A + B + C)(A + B' + C)(A' + B + C) \\ \text{Idempotency} &= (A + B + C)(A + B' + C) \\ &\quad (A + B + C)(A' + B + C) \\ \text{Uniting} &= (A + C)(B + C) \\ &= AB + C\end{aligned}$$

Product-of-sums canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Product-of-sums canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

- $A + B + C$

Product-of-sums canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

- $A + B + C$
- $A + B' + C'$

Product-of-sums canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

- $A + B + C$
- $A + B' + C'$
- $A' + B + C'$

Product-of-sums canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

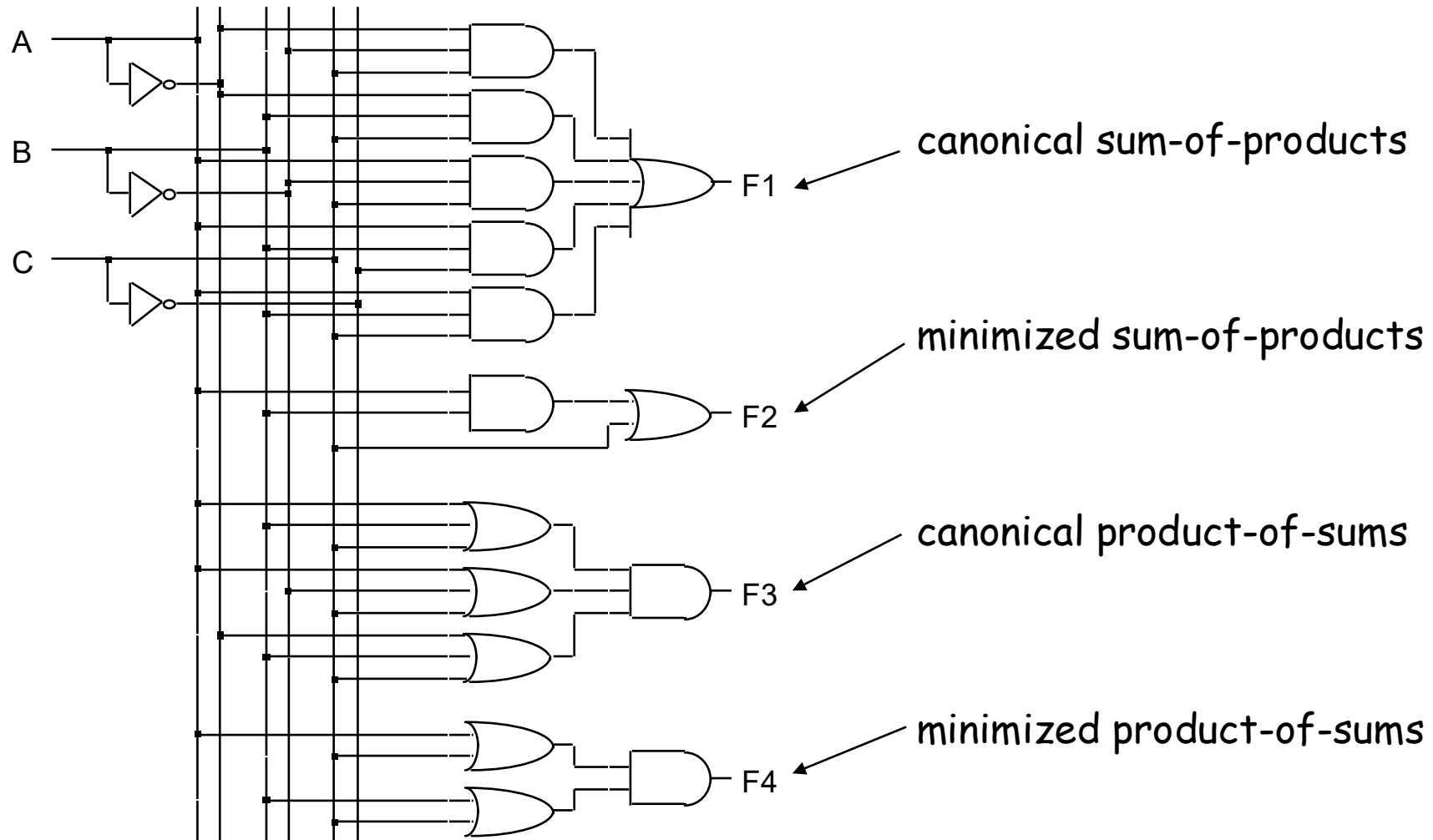
- $A + B + C$
- $A + B' + C'$
- $A' + B + C'$
- $A' + B' + C$

Product-of-sums canonical form

Inputs				Output
A	B	C	D	
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

- $D = (A + B + C)(A + B' + C')(A' + B + C')(A' + B' + C)$

Four alternative two-level implementations of $F = AB + C$



Mapping between canonical forms

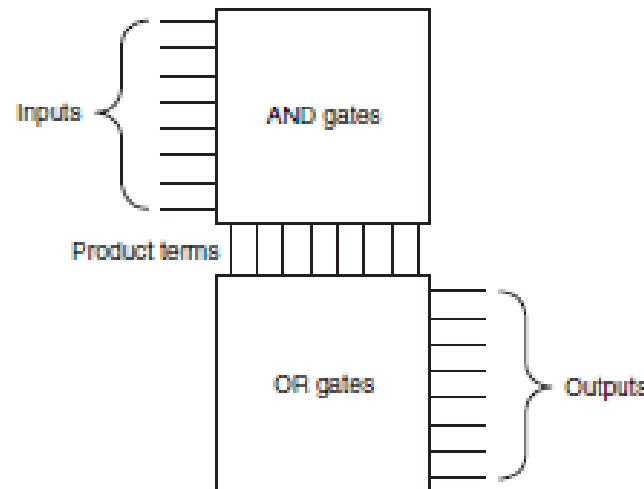
- Minterm to maxterm conversion
 - Use maxterms whose indices do not appear in minterm expansion
 - e.g., $F(A,B,C) = \sum m(1,3,5,6,7) = \prod M(0,2,4)$
- Maxterm to minterm conversion
 - Use minterms whose indices do not appear in maxterm expansion
 - e.g., $F(A,B,C) = \prod M(0,2,4) = \sum m(1,3,5,6,7)$
- Minterm expansion of F to minterm expansion of F'
 - Use minterms whose indices do not appear
 - e.g., $F(A,B,C) = \sum m(1,3,5,6,7)$ $F'(A,B,C) = \sum m(0,2,4)$
- Maxterm expansion of F to maxterm expansion of F'
 - Use maxterms whose indices do not appear
 - e.g., $F(A,B,C) = \prod M(0,2,4)$ $F'(A,B,C) = \prod M(1,3,5,6,7)$

S-o-P, P-o-S, and de Morgan's theorem

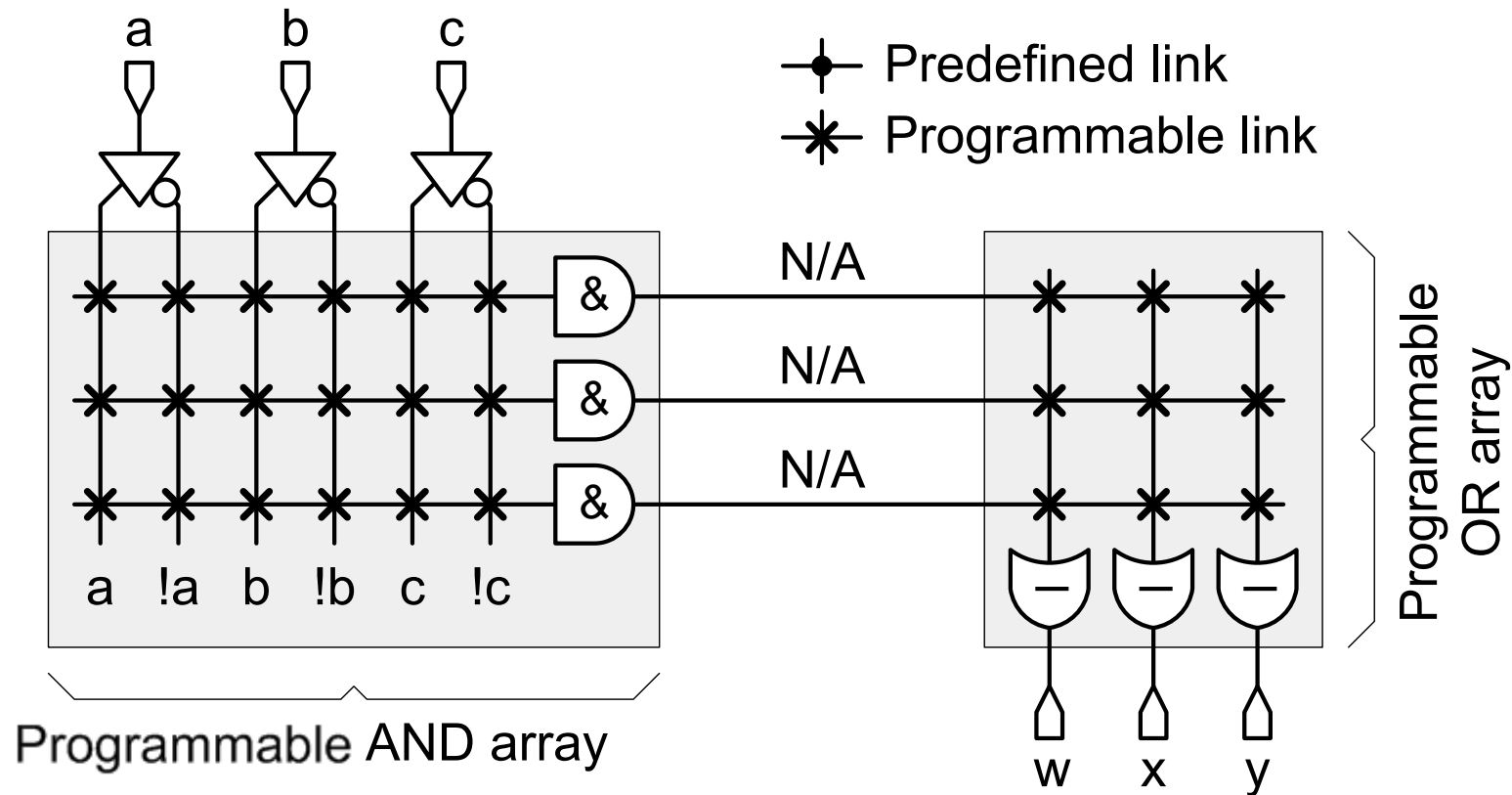
- Sum-of-products
 - $F' = A'B'C' + A'BC' + AB'C'$
- Apply de Morgan's
 - $(F')' = (A'B'C' + A'BC' + AB'C')$
 - $F = (A + B + C) (A + B' + C) (A' + B + C)$
- Product-of-sums
 - $F' = (A + B + C') (A + B' + C') (A' + B + C') (A' + B' + C) (A' + B' + C')$
- Apply de Morgan's
 - $(F')' = ((A + B + C')(A + B' + C')(A' + B + C')(A' + B' + C)(A' + B' + C'))'$
 - $F = A'B'C + A'BC + AB'C + ABC' + ABC$

Programmable Logic Array

- The relationship between a truth table and a two-level representation allows us to generate a gate-level implementation of any set of logic functions.
- The sum-of-products corresponds to a programmable logic array.



Programmable Logic Array



Programmable Logic Array

- Efficient Characteristics
 - only the truth table entries that produce a true value for at least one output have any logic gates associated with them.
 - each different product term will have only one entry in the PLA, even if the product term is used in multiple outputs.

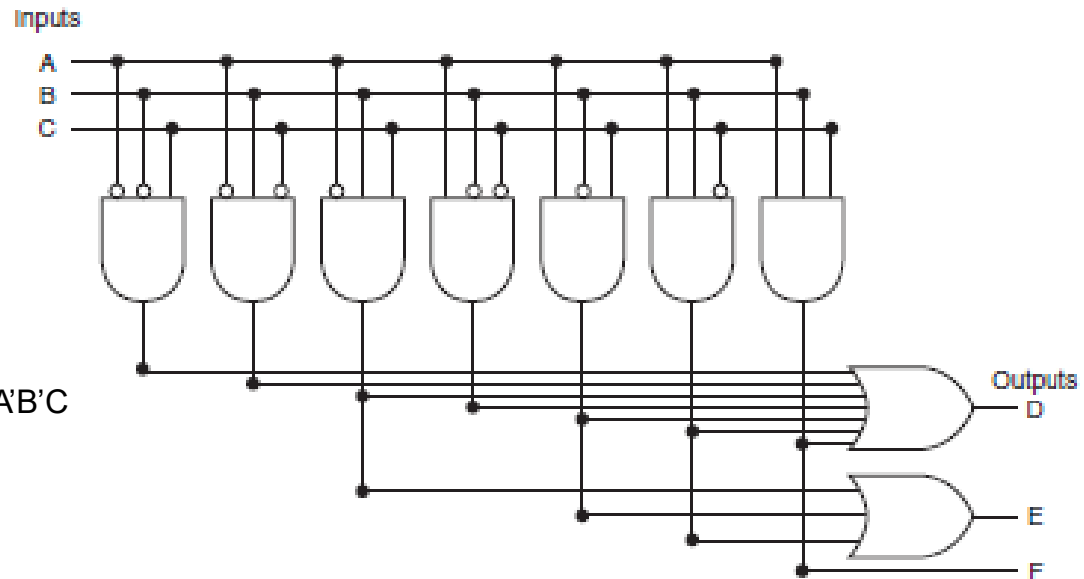
Programmable Logic Array

Inputs				Outputs	
A	B	C	D	E	F
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	0	1

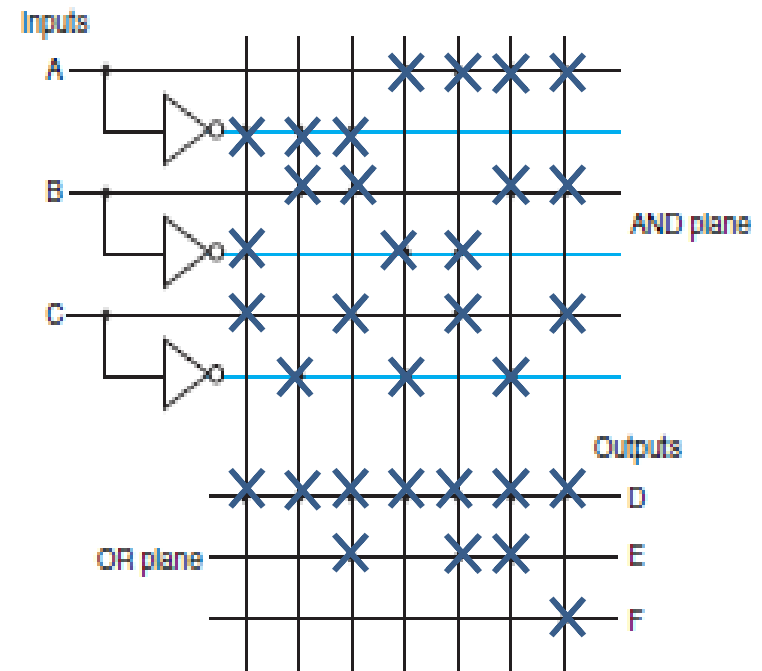
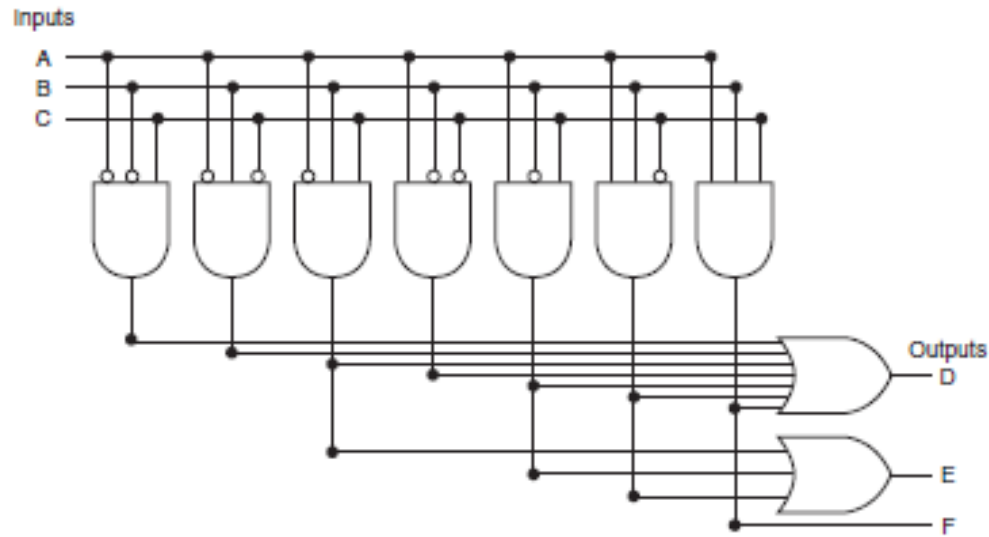
$$D = ABC + ABC' + AB'C + AB'C' + A'BC + A'BC' + A'B'C$$

$$E = A'BC + AB'C + ABC'$$

$$F = ABC$$



Programmable Logic Array



Incompletely specified functions

- Example: binary coded decimal increment by 1
 - BCD digits encode decimal digits 0 – 9 in bit patterns 0000 – 1001

A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

off-set of W

on-set of W

don't care (DC) set of W

these input patterns should never be encountered in practice - "don't care" about associated output values, can be exploited in minimization

Example: Incompletely Specified Functions

Given: Consider the following truth table where F is incompletely specified. Determine the minterm and maxterm expansions for F.

Partial Credit 1: To construct the expansions, we need to include the don't care set along with either the on-set (minterm expansion) or the off-set (maxterm expansion).

Solution 1: The minterm expansion for $F = \sum [m(2, 3, 4) + d(0, 7)]$
The maxterm expansion for $F = \prod [M(1, 5, 6) + D(0, 7)]$

Truth Table			
A	B	C	F
0	0	0	X
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	X



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